Internal Hackathon (SIH 2025)

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- Team Name**:** **LOCORUNNERS**
- Department / Year: **E.C.E - 3rd yr**
- Problem Statement ID: **25020**
- Problem Statement Title: **Development of indigenous contactless Integrated Track Monitoring Systems (ITMS) for Track Recording on Indian Railways**

BRIEF IDEA:

Indigenous Smart Track Monitoring System for Indian Railways

Indian Railways currently depends on costly imported Track Recording Cars (TRCs) and manual inspections for ensuring track safety. These systems are expensive, difficult to customize, and lack standardized traceability of track fittings such as rail clips, liners, and pads, which delays preventive maintenance and reduces accountability.

Our proposed solution is a **low-cost, indigenous, GPS-independent Integrated Track Monitoring System (ITMS)** that combines **hardware and software innovations** while fully supporting modularity and ease of maintenance.

Hardware: A prototype TRC model is developed using ESP32-CAM and MPU6050 sensors to capture vibration and ride-quality data. Each sleeper and fitting is uniquely identified using **laser-based coded markers** or RFID/UHF tags, ensuring precise mapping and lifecycle traceability. An additional **object detection module** detects obstacles such as rocks in landslide-prone zones, enabling a pilot train to automatically reverse and prevent derailments.

Software: All data is transmitted to a central database for real-time processing. Engineers can pinpoint sleeper-level faults at exact location, verifying fitting details (vendor, batch, warranty), and analyse track geometry parameters. Fault detection is **contactless, automated, and compliant with international accuracy standards**.

This system is **scalable, cost-effective, and proactive**, ensuring continuous monitoring, accident prevention, reduced derailments, and complete traceability of track components.

"From Rock to Rail Clip – Nothing Goes Unseen."

"AI on Tracks: See the Fault Before It Happens."